

Honeybee Swarm Dynamics Investigating the Relationship Between Individual Decision Making and Collective Foraging

Javon Hickmon¹, Golnar Gharooni Fard², Orit Peleg^{3,4}

¹ Department of Computer Science, University of Washington | ² Department of Computer Science, University of Colorado Boulder

³ BioFrontiers Institute, University of Colorado Boulder – Boulder, CO | ⁴ Santa Fe Institute - Santa Fe, NM

Correspondence: javonh@cs.washington.edu, orit.peleg@colorado.edu

Abstract

Honeybees are social insects that use pheromone signals to direct each other, resulting in emergent swarm behavior.

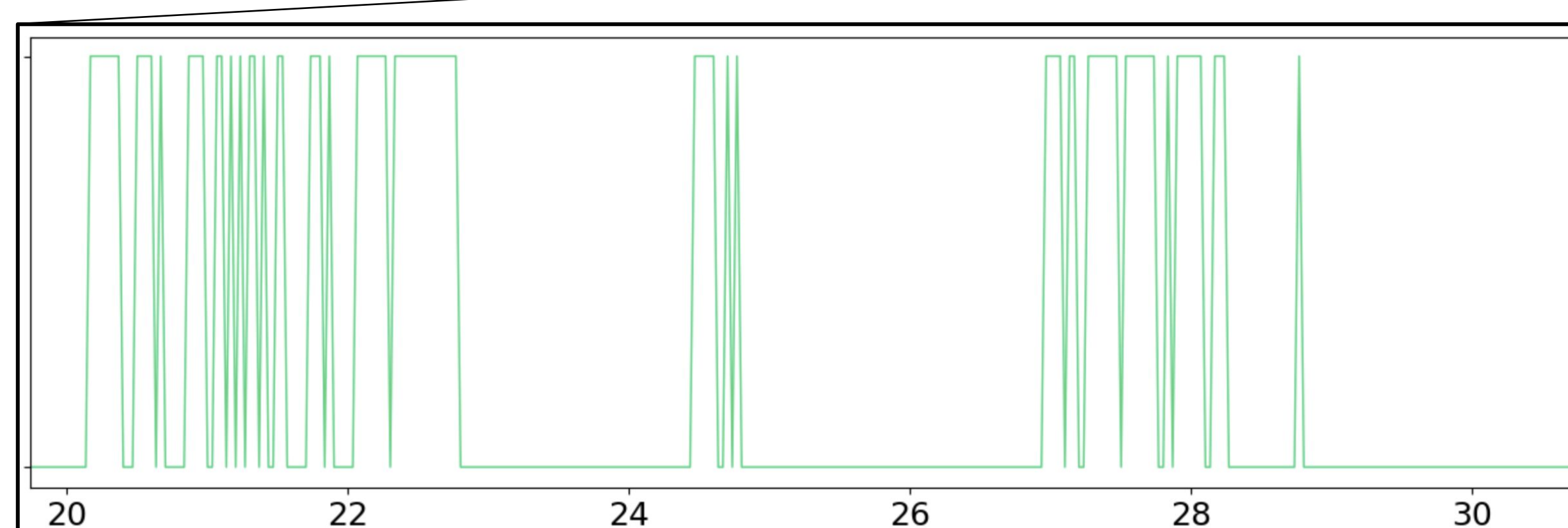
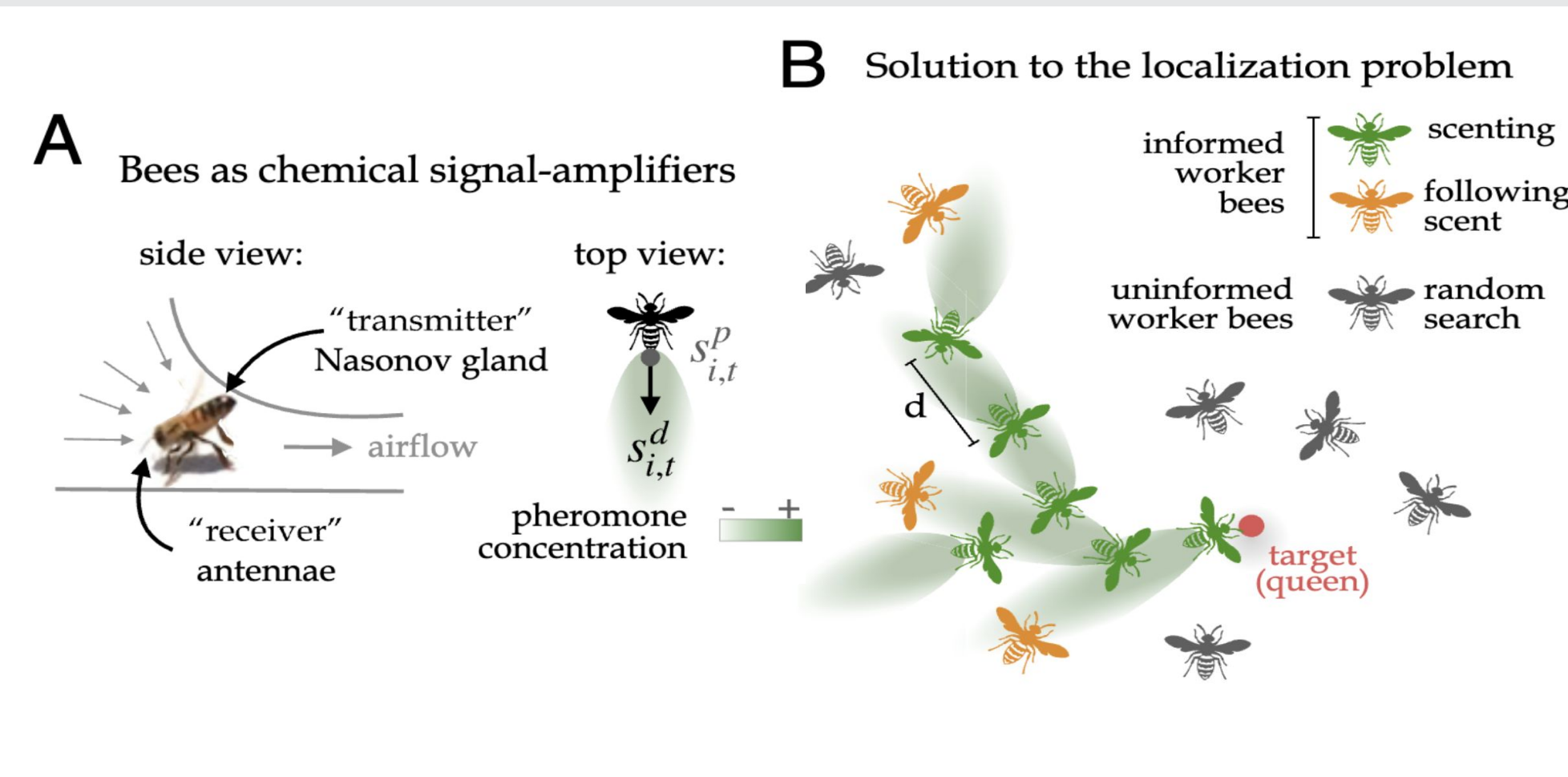
Research has shown that bees can solve the shortest path problem and locate their queen by directing these pheromone signals, or “scenting,” to create a communication network collectively^[2]

Despite this, little work has been done to study how the behavior of the bees changes when in the presence of the swarm.

This study aims to improve the tracking of individual honeybees and investigate how the decisions of individual honeybees affect the foraging and aggregation process of the collective.

Literature Review

- Literature has shown bees collectively create a scenting-mediated communication network by scenting away from the queen (Figure A)^[1]
- Individual bees act as receivers and senders of signals, releasing pheromones from the Nasonov glands and fanning their wings to direct the signals backward (Figure B)^[1]



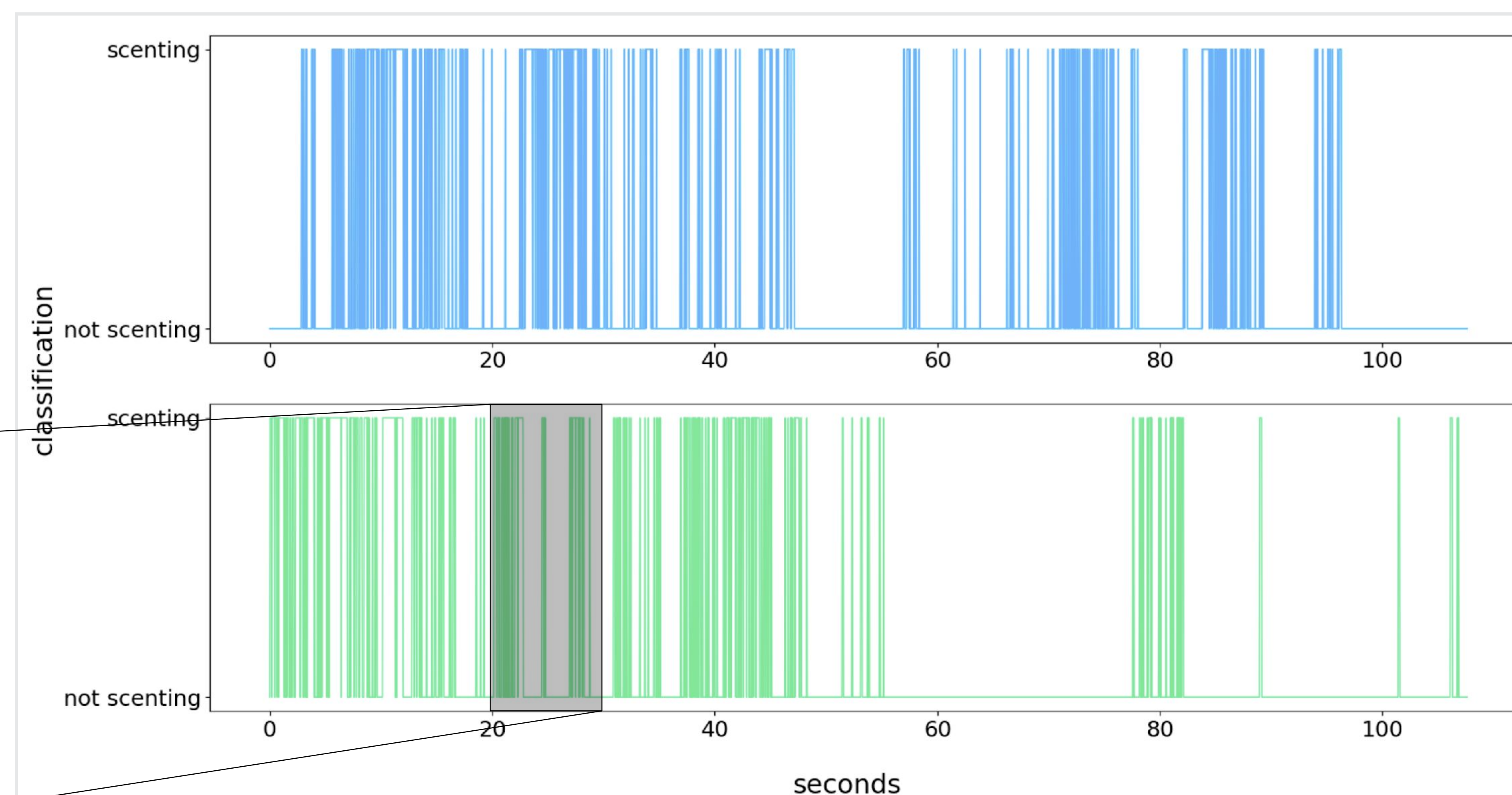
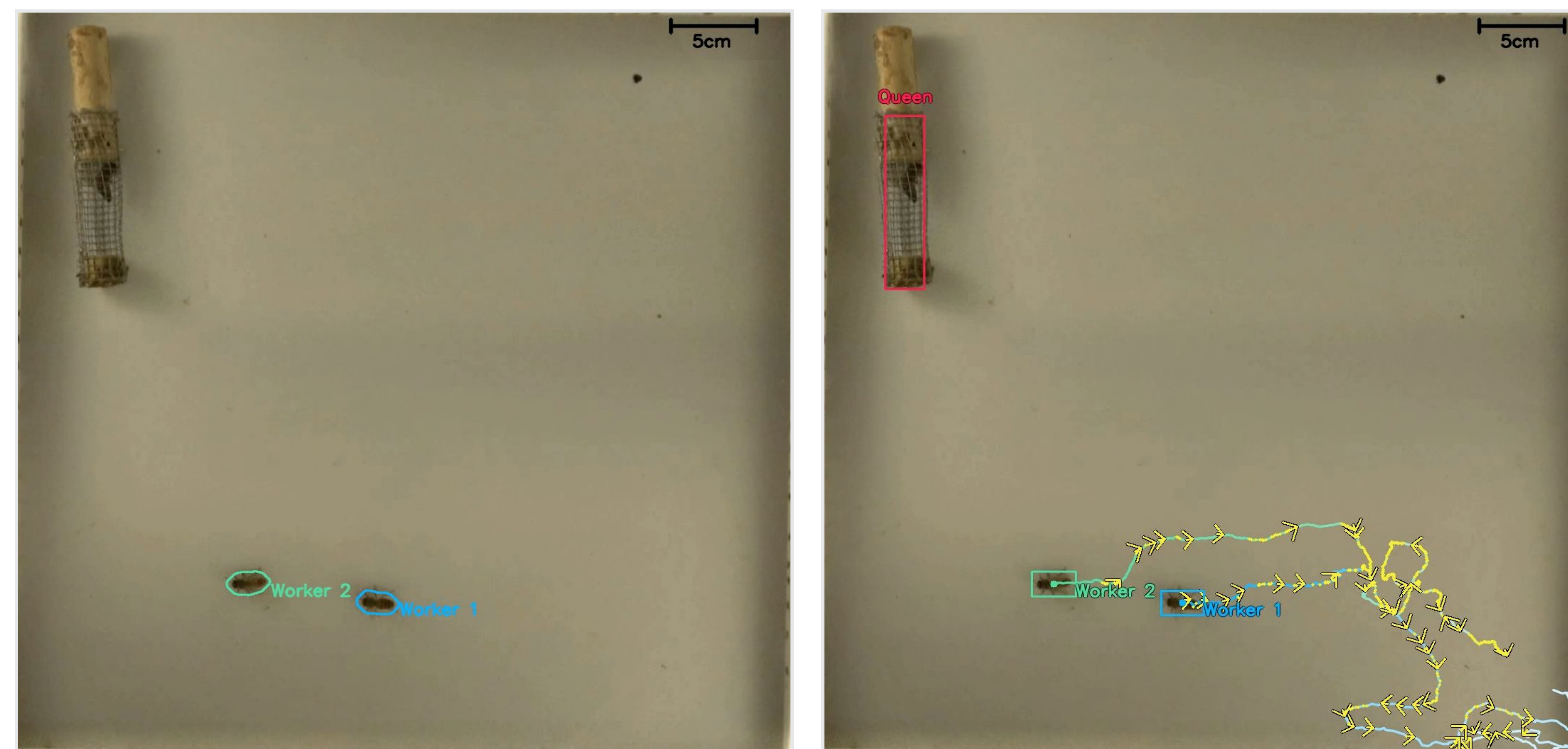
Hypothesis and Research Method

Hypothesis

- We anticipate the time-series data from individual honeybees will display a predictable relationship between scenting and exploring that allows for collective foraging.

Research Methods

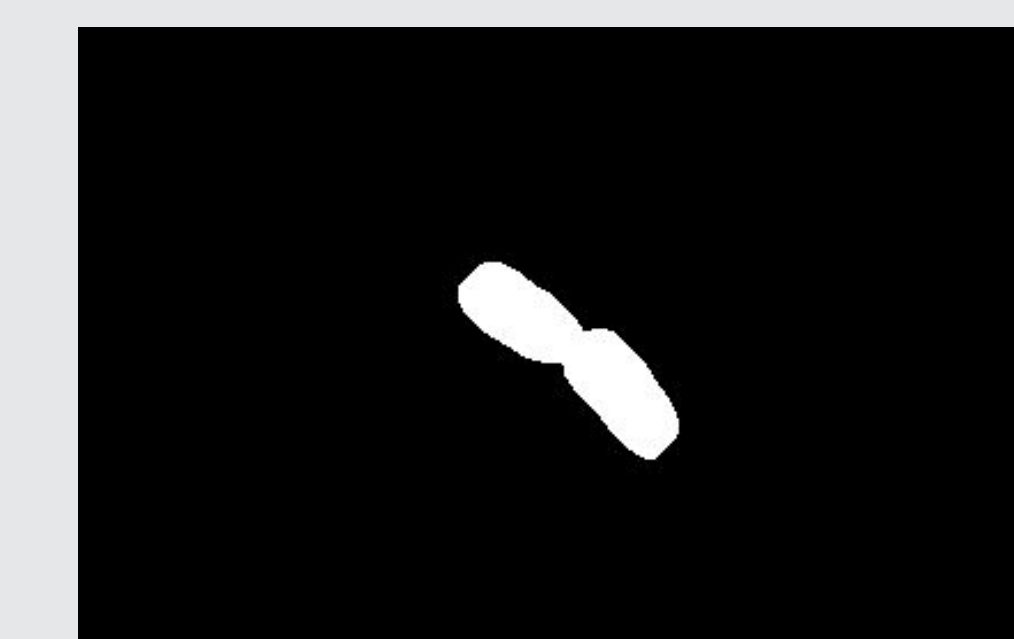
- Established a behavioral assay in which worker bees search for a stationary caged queen in a semi-two-dimensional (2D) arena^[1]
- Created a novel object tracking and segmentation algorithm to gather timeseries data on the scenting behaviors of individual bees when separated from their queen



Our Algorithm

Algorithm 1 Honeybee Tracking and Segmentation

```
background = get_background(video)
for frame ∈ video do
    processed = preprocess(frame, background)
    contours = form_contours(processed)
    tracks = create_tracks(contours, prev_tracks)
    split_tracks = split_groups(tracks)
    frame = draw_tracks(frame, split_tracks)
    show_frame(frame)
    prev_tracks = split_tracks
end for
```



preprocess()



form_contours()



create_tracks()



split_groups()

Next Steps

- Run similar tests with a larger number of bees and improve the tracking algorithm to require less human interaction
- Analyze resulting distributions and determine the correlation between the data for each trial

References

- [1] Nguyen, D. M. T., Iuzzolino, M. L., Mankel, A., Bozek, K., Stephens, G. J., & Peleg, O. (2021). Flow-mediated olfactory communication in honeybee swarms. *Proceedings of the National Academy of Sciences*, 118(13), e2011916118.
- [2] Nguyen, D. M. T., Fard, G. G., Atkins, A., Bontempo, P., Iuzzolino, M. L., & Peleg, O. (2023). Honey bees find the shortest path: a collective flow-mediated approach. *Artificial Life and Robotics*, 28(1), 1-7.